

mance by integrating discrete FPGAs, digital signal processing (DSP) and microprocessor devices into a single, user-customisable SoC.

The built-in hardware IP cores, which were not present earlier, are available now, providing more predictable performance figures. Every year we see shrinking of the process node in line with the industry growth and Moore's Law. Now 45nm, 28nm, 20nm and 16nm FPGAs are available. Also, now we are moving towards 14nm and 10nm nodes. This favours better density and reduction of power.

Adaptive look-up tables

Over the years, the basic building block of the FPGA logic fabric—the look-up table (LUT) based function generator—has remained almost the same. Experimentations were done with the number of inputs for LUT such as three, four, six and eight. Now they have adaptive LUTs which allow two outputs per LUT with two function generators to be implemented, sharing some of the inputs. The FPGAs mainly evolved by providing the same set of resources with different number of resources per family.

In addition, today it is all about providing ecosystems to the designer—evaluation boards, reference designs, design communities and partner IPs, ensuring FPGAs are able to meet the stringent time-to-market goals of the customer.

Things to consider while selecting an FPGA

If we presume that a device meets the basic design requirements, the next four things an engineer needs to consider are power consumption, board space, solution cost and design time. Whatever the application may be, these four factors can bias an engineer's decision across the vendors.

When it comes to low power consumption, iCE40 FPGA family from Lattice Semiconductor and IGL00 series of low-power FPGAs from Microsemi are some great choices in most low-power and battery-operated applications. It is particularly favourable in the Internet of Things (IoT) space where ultra-low-power programmable devices with ultra-small packaging are what a design engineer would be looking for.

Available resources, such as look-up tables (LUTs), registers, I/Os, phase-locked loops (PLLs) and memory, are considered before selecting an FPGA.

Another point to look for is the dedicated hard blocks that could be interfaced with the FPGA, such as the physical coding sublayer (PCS), peripheral component interconnect express (PCIe) end point and double data rate (DDR) memory controller. These could be added to the periphery for high-speed interface with the external world.

If the application requires a processor and an FPGA, then user-customisable SoCs are the best choice.

Selecting FPGAs for an application

High-end FPGAs feature:

- High performance
- High logic density
- High-speed interfaces
- UltraScale and 3D architecture
- Support complex features, architectural blocks and intensive computations
- High power consumption
- High cost
- Example: Virtex family by Xilinx, Stratix family by Altera, ProASIC3 family by Microsemi and Speedster 22i family by Achronix

Low-end FPGAs feature:

- Low power consumption per chip
- Low cost
- Low complexity
- Low speed of operation
- Low logic density
- Low-speed interconnects
- Do not support some features and architectural blocks
- Example: Spartan family by Xilinx, Cyclone family by Altera, Mach X0 and iCE40 families by Lattice Semiconductor and Fusion family by Microsemi

Mid-range FPGAs

The mid-range FPGAs are an optimal solution that provide the users with a balance of cost and performance.

Example: Artix-7 and Kintex-7 series by Xilinx, Arria series by Altera, ECP5/ECP3 series by Lattice Semiconductor and IGL002 series by Microsemi

Selecting FPGAs

Over the past few years, a distinct segmentation of FPGA vendors has occurred depending on the device capabilities: distinct high-end and low-end product lines. If the application requires intensive computation, the FPGAs preferred are those with complex DSP blocks, high-speed inputs/outputs (I/Os) and configurable memories. They are seen serving consumers looking at ultra-high-definition televisions, 3D televisions and medical applications, such as 265-channel ultrasound. Examples of FPGAs delivering these requirements are Virtex, Stratix, ProASIC3 and Speedster 22i.

For low-cost applications, where a trade-off has to be made for perfor-



Lattice ECP3 low-power FPGA

mance, FPGA families, such as Spartan, Cyclone ECP5 and Fusion, are used. Typical applications include emerging communication products, such as heterogeneous networks (HetNet), gigabit passive optical network optical line terminals (GPON OLTs) and small form-factor pluggable (SFP) modules.

Choice of process node

Process nodes are available in 45nm, 40nm, 32nm, 28nm, 22nm, 20nm and 16nm values. The 10nm nodes are already under development.

From the user's perspective, a process node is not everything and the user does not have much control on it. Provided a device meets the design specifications, power and cost budgets and availability, designers often tend to overlook the process node.

However, because of the reduction in process node values, the newer